

Graph Isomorphism with Representatives

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Abstract

In this paper, we bring some insight to the graph isomorphism problem from the field of algebraic graph theory. Given the adjacency matrices, rational canonical bases are utilized because of their invariance property. Those vectors in the bases would reveal the information about the automorphism. Moreover, the bases are extracted from the finite field $\mathbb{Z}_2$, which simplifies the computation significantly. By means of clustering the nonzero entries over $\mathbb{Z}_2$ to find the mapping between those featured vertices, it is interesting to investigate the graph isomorphism problem in this combinatorial point of view. Some strategies based on this formulation especially for graphs with representatives are then proposed.

1 Introduction

Given two undirected graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$, the Graph Isomorphism problem (GI) is to determine whether G_1 and G_2 are isomorphic. In other words, the GI problem is to determine if there exists a bijective function between the two vertex sets V_1 and V_2 such that the adjacency of vertices will be preserved under this function mapping. Geometrically speaking, if two graphs are isomorphic, then they have exactly the same structure regardless of the vertex ordering.

The GI problem has for a long time been a tough question to answer in graph theory [20]. It can be neatly described and seems to give us many intuitions to track down the answer for it. Although there are some algorithms (e.g., [23]) with practically good performance and some other randomized approaches (e.g., [3]), the hardness of the GI problem remains open.

1.1 Computational Complexity of GI

The difficulty of testing isomorphism on graphs was brought up by Garey and Johnson [16] in 1979. It was highlighted in the list of 12 open problems at the end of their book [16]. Up until now, there are only two open problems in the list left pending [18], including GI. In spite of the fact that we cannot prove the computational complexity of GI at this moment, there are some algorithms which work very efficiently for some special graphs, and some others which perform in moderately exponential time (e.g., $\exp(O(\sqrt{n \log n}))$ in [8]). For some particular kinds of graphs, such as trees [2], planar graphs [17], partial k-trees [10], graphs with bounded degree [22], bounded genus [24], bounded eigenvalue multiplicity [6] and so forth, we can determine whether two graphs are isomorphic in polynomial time.

In practice, the graph isomorphism problem can be solved efficiently by less time-consuming algorithms on average. Nauty, designed by McKay [23], is one of the most powerful programs for testing GI; however, it has been shown that there exists a family of graphs which forces McKay's algorithm to run in exponential time [26]. There are also some randomized algorithms for GI. In [3], among those coin-flipping algorithms, Babai first introduced the term "Las Vegas Algorithm" for GI in an attempt to distinguish those probabilistic algorithms whose correctness of the result can be checked from classical Monte-Carlo algorithms (e.g., Strassen-Solovay primality test).

According to related works above, it seems that the solution to GI may not be too hard to find. However, there have been literatures giving supporting evidences showing that GI is located in the low hierarchy in NP [27], which implies that it cannot possibly be NP-complete unless the polynomial time hierarchy collapses to its second level. Since the graph isomorphism problem is neither known to be tractable nor NP-complete for quite a period of time, it has been a good candidate for

an example of problems lying in NP-intermediate (NPI), especially as we know that NPI is not empty if $P \neq NP$ [21] and are eager to find one such instance. In the past few decades, those were once doubted to be in NPI such as Composite Number and Linear Programming, are all proved to be in P [1, 19], whereas GI is still the one to be verified.

The Subgraph Isomorphism Problem is yet another related important problem; it can be viewed as a generalized version of GI. This problem takes two graphs G and H as an input, asking whether G is a subgraph of H . By reducing the Maximum Clique Problem to it, the Subgraph Isomorphism Problem has actually been proved to be an NP-complete problem [11]. Let $G = K_n$, where K_n is the complete graph of size n , for $n = 1, \dots, |H|$. If we can decide whether G is a subgraph of H in polynomial time, then we can also find out the maximum clique size in polynomial time.

1.2 An Overview of Our Approach

Given the adjacency matrix A_G and A_H of the graph G and H , respectively, our strategy is to try to discover whether it is possible to set up a “canonical invariant matrix” for both A_G and A_H , which contains vectors in the rational canonical basis in its columns.

The failure of the early development on constructing a polynomial time algorithm for GI directly based upon techniques from graph theory urges researchers to turn to investigate it with other mathematical tools, such as group theory and topology. For example, Babai and Luks proposed an algebraic approach for assigning canonical forms to general graphs in moderately exponential time [8]. In fact, canonical labeling algorithms can have good performance on average, that is, some of those strategies certainly behave well on almost all graphs [7, 5]; the hard case is when G is a strongly regular graph [4], which tends to have a large number of automorphisms in many cases. Analogously, the approach in this paper is to categorize vertices into groups by applying some fundamental theorems about the adjacency matrix from algebraic graph theory. It is the concept of permuting the canonical basis corresponding to each vertex that we use for classification.

Our attempt is to control the mapping on a particular reduced set of nodes, which is called a set of *representatives*, therefore it is more applicable when we already have some prior knowledge for classifying some of the nodes. For those graphs

with representatives, more details can be found in Section 4.1.

The only preprocessing is to compute the *rational canonical form* of the adjacency matrices over the Galois field with characteristic 2, which is often denoted as $\mathbb{Z}_2$ or $GF(2)$, and then we can find a permutation on nodes by determining a *rational canonical basis* heuristically. One thing to notice is that, to the best of our knowledge, all the related works before in algebraic graph theory relies on the calculation over an infinite field (mostly over $\mathbb{R}$ or $\mathbb{C}$), therefore computing those numerical parts requires sufficiently high precision, for otherwise the approximation error incurred when computing the spectral system may lead to an unacceptable incorrect result. Since all the matrix operations here are over $\mathbb{Z}_2$, where addition is exclusive-OR (XOR) and multiplication is the AND operation, our algorithm only requires simple arithmetic to sketch the essence of those fundamental theories.

2 Preliminaries

In this section, we provide some background knowledge needed in the domain of linear algebra.

2.1 Notations and Basic Linear Algebra

Since we want to ensure the existence of the bijection, by the definition of GI, permutations are always what we concern. In abstract algebra, these permutations are maintained by the symmetric group and its substructure. A symmetric group G , denoted by $Sym(M)$, is a finite group whose elements are permutations of a given set M , and whose group operation is the composition of permutations in $Sym(M)$. The permutation group, denoted by S_n , is a subgroup of the symmetric group $Sym(M)$ which consists of all permutations on n symbols. Clearly, the symmetric group S_n is of order $n!$. The relation that two graphs G and H are isomorphic, denoted as $G \cong H$, can also be perceived in the automorphism point of view. Let $\sigma(G)$ denote the graph obtained from G by applying the permutation σ . The set collecting all automorphisms on a graph G is defined as $Aut(G) = \{\sigma \in Sym(V_G) \mid \sigma(x) \cong G\}$. Permutation matrices are what we use to arrange the ordering in matrix algebra. A permutation matrix P is a square matrix of size n that has exactly one entry 1 in each row and each column and 0's elsewhere. Every permutation matrix P is an

orthogonal matrix, i.e., $P^T P = I$, where I is an identity matrix.

Let A be an $n \times n$ square matrix. If there is a nonzero vector $x \in \mathbb{R}^n$ such that $Ax = \lambda x$ for some scalar λ , then λ is called the eigenvalue of A with corresponding eigenvector x (or x is the eigenvector of A with respect to λ). The spectrum of A , denoted as $\lambda(A)$, is the multiset of its eigenvalues. The union of the zero vector and the set of all eigenvectors with respect to an eigenvalue λ forms a subspace of $\mathbb{R}^n$, which is known as the eigenspace of λ , denoted by $V(\lambda)$. Let $Ker(A)$ be the kernel space (or null space) of a matrix A . By definition we know that $V(\lambda) = \{x \in \mathbb{R}^n \mid Ax = \lambda x\} = Ker(A - \lambda I)$. The algebraic multiplicity of an eigenvalue λ is defined as the multiplicity of the root λ of the characteristic polynomial, whereas the geometric multiplicity of λ is the dimension of the eigenspace $Ker(A - \lambda I)$, often denoted as $nullity(A - \lambda I)$. The multiplicity of the one having the maximum multiplicity among all the eigenvalues of A is defined as the *multiplicity* of A . Suppose there is a vector space V and a linear transformation T , a subspace $W \subseteq V$ is said to be *T-invariant* if for all $x \in W$, $T(x) \in W$. Indeed, $V(\lambda)$ is an invariant subspace of $\mathbb{R}^n$ under A , while A is regarded as a linear transformation.

Given two square matrices A and B , if there exists an invertible matrix Q such that $Q^{-1}AQ = B$, then we would say that A is similar to B . In particular, if there exists a diagonal matrix B such that A is similar to B , then we say A is diagonalizable. Also, if there exists an orthogonal matrix Q such that $Q^{-1}AQ = \Lambda$, we would say that A is orthogonally diagonalizable. A well-known fact is that A is diagonalizable if and only if there exists n linearly independent eigenvectors of A , i.e., there exists an invertible matrix Q having these eigenvectors as its columns so that $Q^{-1}AQ = \Lambda$, where Λ is a diagonal matrix with eigenvalues of A as its diagonal entries.

2.2 Eigenspaces and Automorphisms

Given a graph $G = (V_G, E_G)$, suppose all the vertices in G are labeled from 1 to n , where $n = |V_G|$. For an adjacency matrix A_G of G , if we want to interchange the label of two vertices i and j , we can simply multiply the permutation matrix P_π corresponding to the permutation π at both sides of A_G to get the adjusted adjacency matrix for the new labeled graph, i.e. $P_\sigma A P_\sigma$, where $\sigma(i) = j$, $\sigma(j) = i$ and $\sigma(k) = k$ for all $k \neq i, j$. We merely exchange the i th and the j th

row when multiplying to the left and the i th and the j th column to the right without altering the adjacency relation in between. This special kind of permutation is called the transposition. Notice that if applying the same operation again, vertex i and vertex j change back to its original position, i.e. $P_\sigma^2 = I \Rightarrow P_\sigma^{-1} = P_\sigma$, thus $P_\sigma^T = P_\sigma$. Since every permutation can be factored into a product of transpositions, for any permutation $\pi \in S_n$, we have

$$\begin{aligned} A_{\pi(G)} &= (P_{\sigma_k} \dots (P_{\sigma_2} (P_{\sigma_1} A_G P_{\sigma_1}) P_{\sigma_2}) \dots P_{\sigma_k}) \\ &= (P_{\sigma_1}^T P_{\sigma_2}^T \dots P_{\sigma_k}^T)^T A_G (P_{\sigma_1} P_{\sigma_2} \dots P_{\sigma_k}) \\ &= (P_{\sigma_1} P_{\sigma_2} \dots P_{\sigma_k})^T A_G (P_{\sigma_1} P_{\sigma_2} \dots P_{\sigma_k}) \\ &= P_\pi^T A_G P_\pi, \end{aligned}$$

where $\sigma_i \in S_n$ is the permutation of exchanging exactly two positions on a set, and $P_\pi = P_{\sigma_1} P_{\sigma_2} \dots P_{\sigma_k}$, for some $k \in \mathbb{N}$. This results in the fact that $A_{\pi(G)}$ is orthogonally similar to A . Furthermore, since $A_{\pi(G)}$ and A_G are both real symmetric, there exists an orthogonal matrix Q_{A_G} such that $A_G = Q_{A_G} \Lambda_{A_G} Q_{A_G}^T$ where Λ_{A_G} is a diagonal matrix with all eigenvalues of A_G on its diagonal, thus $A_{\pi(G)} = P_\pi^T (Q_{A_G} \Lambda_{A_G} Q_{A_G}^T) P_\pi = (Q_{A_G}^T P_\pi)^T \Lambda_{A_G} (Q_{A_G}^T P_\pi)$. Since $\Lambda_{A_G} = \Lambda_{A_{\pi(G)}}$ and $Q_{A_G}^T P_\pi$ is an orthogonal matrix, we conclude that the columns of $P_\pi^T Q_{A_G}$ reveals n linearly independent eigenvectors of $A_{\pi(G)}$. Due to the fact that A_G and P_π are commutative under multiplication if and only if π is an automorphism on G , the deductions above can also be derived from the following well-known result [6] in the spectral graph theory.

Proposition 1. *Let G be an undirected graph with an $n \times n$ adjacency matrix A . A permutation $\pi \in Aut(G)$ if and only if the eigenspaces of A are invariant under P_π .*

3 Rational Canonical Bases and Graph Automorphisms

Similar to the idea of invariant eigenspaces, we will briefly review the annihilating polynomial and yet another shape of the invariant kernel spaces other than the generalized eigenspaces while composing a rational form, and then describe the strategies applied in this section.

To achieve the simplicity on numerical computation, all of the matrices here are manipulated

over $\mathbb{Z}_2$ as we stated before, since adjacency matrices are all boolean matrices. There has been quite some work done related to the finite field linear algebra, such as finding the null space from the reduced row echelon form (RREF) and matrix multiplication; more details for these can be found in [9]. We next show the adequacy of such an arrangement.

3.1 Invariance of Rational Canonical Bases

The invariant eigenspaces urge us to have the same strategy in the field $\mathbb{Z}_2$, whereas the major problem is that a characteristic polynomial might not be able to split into linear factors. Therefore, we generalize this idea to the kernel spaces emerged from which we usually called the *rational canonical form* (or *Frobenius normal form*). To bring up the next proposition, background knowledge about the rational form is provided here in brief; one can refer to [14] for more explicit details.

Let $V = \mathbb{F}^n$ and $f(t) = (-1)^n(\phi_1(t))^{n_1}(\phi_2(t))^{n_2}\dots(\phi_k(t))^{n_k}$ be the characteristic polynomial of a matrix $A \in M_{n \times n}(\mathbb{F})$, where the $\phi_i(t)$'s are distinct irreducible monic polynomials and the n_i 's are positive integers, for $1 \leq i \leq k$. Let v be a nonzero vector in V . A polynomial $p(t)$ is the *annihilator* of v if $p(t)$ is a monic polynomial of least degree for which $p(A)(v) = 0$. Let $K_{\phi_i} = \{u \in V \mid (\phi_i(A))^p(u) = 0 \text{ for some positive integer } p\}$. It can be shown that each K_{ϕ_i} is a nonzero A -invariant subspace of V . The *companion matrix* of a monic polynomial $q(t) = c_0 + c_1t + \dots + c_{k-1}t^{k-1} + t^k$ is the square matrix

$$X = \begin{pmatrix} 0 & 0 & \cdots & 0 & -c_0 \\ 1 & 0 & \cdots & 0 & -c_1 \\ 0 & 1 & \cdots & 0 & -c_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & -c_{k-1} \end{pmatrix}.$$

The objective here is to build up an invertible Q such that $Q^{-1}AQ = C$, and C is of the form

$$\begin{pmatrix} C_1 & 0 & \cdots & 0 \\ 0 & C_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & C_r \end{pmatrix},$$

where each C_i is the companion matrix of a polynomial $(\phi(t))^m$ such that $\phi(t)$ is a monic irreducible divisor of the characteristic polynomial of

A and m is a positive integer for $i = 1, \dots, r$. We name the set of those n column vectors in Q a *rational canonical basis* for A .

The smallest A -invariant subspace which contains a vector $v \in V$ is the *A-cyclic subspace* $\beta_v = \text{span}\{v, A(v), A^2(v), \dots\}$ generated by v . Suppose that $p(t) = (\phi_1(t))^{m_1}(\phi_2(t))^{m_2}\dots(\phi_k(t))^{m_k}$ is the minimal polynomial of A , then it is known that $K_{\phi_i} = \text{Ker}((\phi_i(A))^{m_i})$, $\dim(K_{\phi_i}) = d_i n_i$ for each i , where d_i is the degree of $\phi_i(t)$, and K_{ϕ_i} has a basis consisting of the union of A -cyclic bases. In the following two propositions, we show the invariance properties of these null spaces.

Proposition 2. Suppose π is an isomorphism from G to H and $P_\pi^T(v_i) = u_i$, where β_{v_i} is an A_G -cyclic subspace and β_{u_i} is an A_H -cyclic subspace, then $P_\pi^T A_G^k(v_i) = A_H^k(u_i)$, for all $k \in \mathbb{N}$.

Proof. Since $P_\pi^T A_G P_\pi = A_H$, $P_\pi^T A_G^k(v_i) = P_\pi^T P_\pi A_H^k P_\pi^T(v_i) = A_H^k(u_i)$. $\square$

Proposition 3. Suppose A is the adjacency matrix of G with characteristic polynomial $f(t) = (-1)^n(\phi_1(t))^{n_1}(\phi_2(t))^{n_2}\dots(\phi_k(t))^{n_k}$. If π is a permutation that belongs to $\text{Aut}(G)$, then K_{ϕ_i} is invariant under P_π , for all $i \in \{1, 2, \dots, k\}$.

Proof. For any $x \in K_{\phi_i}$, suppose that $(\phi_i(A))^p(x) = 0$, for some positive integer p . Then $(\phi_i(A))^p P_\pi(x) = P_\pi(\phi_i(A))^p(x) = P_\pi(0) = 0$, since P_π is commutable with both the power of A and the diagonal matrix. Therefore, K_{ϕ_i} is a P_π -invariant subspace. $\square$

It is also true that every matrix in a finite-dimensional vector space has a rational canonical basis and a unique rational canonical form, although the rational canonical bases are not. The primary decomposition theorem for rational canonical form states that V is the direct sum of $K_{\phi_1}, K_{\phi_2}, \dots, K_{\phi_k}$, i.e., $V = K_{\phi_1} \oplus K_{\phi_2} \oplus \dots \oplus K_{\phi_k}$, and if A_i is the restriction of A to K_{ϕ_i} and C_i is the rational canonical form of A_i for $1 \leq i \leq k$, then $C = C_1 \oplus C_2 \oplus \dots \oplus C_k$ is the rational canonical form of A . A consequence of the decomposition theorem is that a rational canonical basis as a shape of the unions of cyclic subspaces can be formed. By examining the invariance property of these cyclic subspaces, we next show that it is sufficient to demonstrate those automorphisms in graphs.

Theorem 1. Let G be a graph with adjacency matrix A . A permutation $\pi \in \text{Aut}(G)$ if and only if the subspaces $K_{\phi_1}, K_{\phi_2}, \dots, K_{\phi_k}$ are invariant under P_π .

Proof. For any $x \in K_{\phi_i}$, $i \in \{1, 2, \dots, k\}$, $(\phi_i(A))^{m_i} P_\pi(x) = P_\pi(\phi_i(A))^{m_i}(x) = P_\pi(0) = 0$, hence $P_\pi(x) \in K_{\phi_i}$. Now suppose that each K_{ϕ_i} is P_π -invariant. Let Q be an invertible matrix with each column being selected from K_{ϕ_i} for some i , such that $Q^{-1}AQ = C$, where C is the rational canonical form of A . Then $C = (P_\pi Q)^{-1}A(P_\pi Q) = Q^{-1}P_\pi^T A P_\pi Q$, which implies that $P_\pi^T A P_\pi = A$. Therefore, $AP_\pi = P_\pi A$. $\square$

3.2 The Algorithm

In our strategy, we intend to seek for those *representatives* in graphs. Gaining insight of the canonical basis structure is the prerequisite for finding representatives.

The *dot diagram* is often used to display the structure of unions of cyclic bases. Given a rational canonical basis β and an irreducible monic divisor $\phi(t)$ of the characteristic polynomial of A , let $\beta_{v_1}, \beta_{v_2}, \dots, \beta_{v_k}$ be the cyclic bases of β that are contained in K_ϕ , and $(\phi(t))^{p_j}$ be the annihilator of v_j for each j . The cyclic bases are arranged in decreasing order of size so that $p_1 \geq p_2 \geq \dots \geq p_k$ and every β_{v_j} contains dp_j vectors, where d is the degree of $\phi(t)$. Here we define the dot diagram of $\phi(t)$ to be the array consisting of k rows of dots from top to bottom, with p_j dots in the j th row; each row begins from the right and terminates after p_j dots. According to Proposition 2, since $\beta_{v_i} = \{v_i, A(v_i), A^2(v_i), \dots, A^{dp_i-1}(v_i)\}$ is well-specified whenever v_i is determined, those leftmost points in the dot diagram of $\phi(t)$ play an extremely important role; that is to say, finding a mapping from v_i to u_i is sufficient for β_{v_i} and β_{u_i} . While there are k rows in the dot diagram which is also equal to $\text{nullity}(\phi(A_G))$, there are k vectors to be determined. In our design, we first set up a rational canonical basis for the adjacency matrix A_G , and then find out the appropriate $\{v_1, v_2, \dots, v_k\}$ in each K_{ϕ_i} of A_H by comparing with those of A_G .

The procedure of selecting $\{v_1, v_2, \dots, v_k\}$ for K_{ϕ_i} involves in the computation of the null space, which can be obtained by selecting a basis from the reduced row echelon form. The representatives that we seek for are exactly those “free variables” which can be found in RREF during calculating the null spaces. Let $p_0 = p_1 + 1, p_{k+1} = 0$ and $\mathcal{P} = \{\mathcal{P}_1, \mathcal{P}_2, \dots, \mathcal{P}_t\}$ be the partition of $\{v_1, v_2, \dots, v_k\}$ such that each cell $\mathcal{P}_j$ in $\mathcal{P}$ is the set $\{v_l, v_{l+1}, \dots, v_{l+r}\}$ where $p_{l-1} > p_l = p_{l+1} = \dots = p_{l+r} > p_{l+r+1}$, for some $l \in \{1, \dots, k\}$, $r \geq 0$, and $1 \leq t \leq m_i$. Let $r_j(\phi_i)$ denote a

set of indices of the free variables while computing $\mathcal{P}_j$, i.e., $|r_j(\phi_i)| = r + 1$ and each $v_k \in \mathcal{P}_j$ is a cyclic vector obtained by setting exactly one variable in $r_j(\phi_i)$ to be 1 and others to be 0. Let $R(\phi_i) = \{r_1(\phi_i), r_2(\phi_i), \dots, r_t(\phi_i)\}$. Apparently, $R(\phi_i)$ is not unique; we denote the set which contains all the possible $R(\phi_i)$ by $Rpt(\phi_i)$. In fact, this is the main difficulty to decide whether two graphs are isomorphic or not, as the number of distinct $R(\phi_i)$ do intuitively dominate the number of automorphisms to some extent. If we can find the appropriate mapping of the vertices corresponding to the indices in $r_j(\phi_i)$ between G and H for all j , i.e., a bijection on the subset $r_j(\phi_i)$ of G and H , the isomorphism on the whole graph will emerge spontaneously via such a small part.

Consider the extreme case where the input is a complete graph with n vertices (K_n). Suppose if n is odd, the characteristic polynomial of A_{K_n} would be $f(t) = (-1)^n(\phi_1(t))(\phi_2(t))^{n-1}$ where $\phi_1(t) = t$ and $\phi_2(t) = t - 1$, as we can find a unique nonzero vector x such that $Ax = 0$ and $\text{nullity}(A - I) = n - 1$. Therefore each $R(\phi_1) \in Rpt(\phi_1)$ and $R(\phi_2) \in Rpt(\phi_2)$ only contains one set $r_1(\phi_1)$ and $r_1(\phi_2)$ in the partition respectively; moreover, $Rpt(\phi_1) = \{\{\{1\}\}, \{\{2\}\}, \dots, \{\{n\}\}\}$ and $R(\phi_2) = \{\{2, 3, \dots, n\}\} \in Rpt(\phi_2)$ is one of the valid combinations of representatives. That is,

$$\begin{aligned} & \beta_{v_1} \cup \beta_{v_2} \cup \dots \cup \beta_{v_{n-1}} \\ &= \{v_1, v_2, \dots, v_{n-1}\} \\ &= \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \right\} \end{aligned}$$

is one of the bases for K_{ϕ_2} . Let $Q = (x \ v_1 \ v_2 \ \dots \ v_{n-1})$, then $Q^{-1}A_{K_n}Q$ will be the canonical form. Furthermore, no matter how we permute those rightmost $n - 1$ column vectors in Q , this new-formed matrix Q' resembles a certificate of the automorphism, since there always exists a permutation matrix P_π such that $Q' = P_\pi Q$, and π is one of the automorphisms on K_n . An observation on the counting problem is that $|Aut(K_n)| = n!$ can also be inferred here due to the fact that $|Rpt(\phi_2)| = n$ and the number of valid column permutations is $(n - 1)!$. If n is even, then $f(t) = (-1)^n(\phi_1(t))^n$ where $\phi_1(t) = t - 1$ since $\text{nullity}(A - I) = n - 1$ and $\text{nullity}((A - I)^2) = n$, a similar result can be deduced analogously.

Given a rational canonical basis of A_G , our goal here is to find a suitable set of variables for A_H

which provides us the appropriate $\{v_1, v_2, \dots, v_k\}$ in each K_{ϕ_i} of A_H . For a random graph G with small $|Aut(G)|$, this approach works efficiently. Although there are 2^m vectors in a kernel space of dimension m , and still we have to take permutations of those vectors into account for a single K_{ϕ_i} , there is a positive correlation between the maximum dimension of the null spaces and the number of automorphisms in some sense. In other words, the number of automorphisms is partly bounded by the maximum dimension. This enhances the probability to guess the right mapping while dealing with graphs with large $nullity(\phi_i(A_G))$, such as strongly regular graphs.

Initially the set of columns of Q_{A_H} is empty. Each time after deciding a cyclic vector v_j in K_{ϕ_i} , we update Q_{A_H} by appending those p_j vectors in β_{v_j} to it. Notice that for any updated $n \times r$ matrix Q_{A_H} , the permutation π is well-determined, once if every row happens to be unique in the updated Q_{A_H} and if there exists a permutation matrix P_π such that $P_\pi^T Q_{A_G}(:, 1:r) = Q_{A_H}$ where $Q_{A_G}(:, 1:r)$, in the notation of MATLAB, stands for the submatrix of Q_{A_G} forming by the preceding r columns in it. This enables us to rapidly filter impossible arrangements. On the other hand, classifying those same rows into a cluster after updating is the refinement process for the next round, in that we can determine those representative variables based on that partition on the rows. Formally speaking, let $\mathcal{T}_G^i = \{\mathcal{T}_{G_1}^i, \mathcal{T}_{G_2}^i, \dots, \mathcal{T}_{G_q}^i\}$ and $\mathcal{T}_H^i = \{\mathcal{T}_{H_1}^i, \mathcal{T}_{H_2}^i, \dots, \mathcal{T}_{H_q}^i\}$ be the partition on row indices of A_G and A_H , respectively, after retrieving independent vectors from $K_{\phi_1}, \dots, K_{\phi_i}$ for $1 \leq i \leq k$. That is, the j th row is equal to the k th row of Q_{A_G} if and only if both j and k belongs to $\mathcal{T}_{G_t}^i$ for some t ; it is the same as for H . Let $f: \mathcal{T}_G^i \rightarrow \mathcal{T}_H^i$ be a set-mapping bijection such that $f(X) = Y$ if the x th row of Q_{A_G} is equal to the y th row of Q_{A_H} for all $x \in X$, $y \in Y$, and $|X| = |Y|$. We should go back to find another possible $R(\phi_i)$ for A_H if there exists no such f . Thus if the set of representatives $R_G(\phi_{i+1}) = \{r_1(\phi_{i+1}), \dots, r_t(\phi_{i+1})\} \in Rpt(\phi_{i+1})$ is selected for $K_{\phi_{i+1}}$ of A_G , then $R_H(\phi_{i+1}) = \{g(r_1(\phi_{i+1})), \dots, g(r_t(\phi_{i+1}))\}$ is the one for $K_{\phi_{i+1}}$ of A_H , where $g(Z) = \{g(x) \mid x \in Z\}$ and g is the mapping satisfying that, for any $x \in r_l(\phi_{i+1}) \in R_G(\phi_{i+1})$, $x \in Z \in \mathcal{T}_G^i$ if and only if $g(x) \in f(Z)$, for all l .

As a result, the refinement on rows is the process that we step-by-step limit the mapping of the vertices between G and H to eventually obtain the final f as an isomorphism. Figure 1 shows

the overall structure of the algorithm. By applying other design strategies to the third line of the algorithm such as the ones we mention below, our heuristic scheme would become even more adapted.

procedure GI-TESTING(G, H)

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1 Compute  $R(\phi_1), R(\phi_2), \dots, R(\phi_k)$  randomly for
   $A_G$  to construct  $Q_{A_G}$ 
2 for  $i := 1$  to  $k$  do
3   Find the right  $R(\phi_i)$  to form a basis for
     $K_{\phi_i}$  of  $A_H$  according to  $\mathcal{T}_G^{i-1}$  and  $\mathcal{T}_H^{i-1}$ 
4   Append columns to  $Q_{A_H}$  in order from the
    basis for  $K_{\phi_i}$ 
5   Compute the refinement  $\mathcal{T}_G^i$  of  $\mathcal{T}_G^{i-1}$  and  $\mathcal{T}_H^i$ 
    of  $\mathcal{T}_H^{i-1}$ 
6   if  $|\mathcal{T}_H^i| = n$ 
7      $r :=$  the number of columns in  $Q_{A_H}$ 
8     Compute  $P_\pi$  such that  $P_\pi^T Q_{A_G}(:, 1:r)$  is
      equal to  $Q_{A_H}$ 
9     Construct the complete  $Q_{A_H}$  according to  $P_\pi$ 
10    if  $P_\pi^T Q_{A_G} = Q_{A_H}$ 
11      return  $\pi$ 
12    end if
13  end if
14 end for
```

Figure 1: The algorithm that returns an isomorphism from G to H if there exists any.

3.2.1 Hamming Weight and Hamming Distance

Due to the increment of the number of variable combinations while the dimension increases, it is reasonable to start selecting variables of a polynomial ϕ_i with the least $nullity(\phi_i(A_H))$. In order to prune out those inappropriate selections of representatives, we can order the independent vectors found by their Hamming weights to prune out wrong arrangements; similarly, the selection of $R_H(\phi_i)$ for each i takes the advantages of it and also of Hamming distance.

3.2.2 Auxiliary Data Structures

In addition to the numerical simplicity caused by the boolean property, another benefit of $\mathbb{Z}_2$ field arithmetic is to realize the concept of choosing a vertex by setting its corresponding variable to be 1 in a more combinatorial sense, therefore this characteristic involves how to process this type of combinatorial relation well. The *ordered binary deci-*

sion diagram (OBDD) is a succinct data structure which can uniquely represent a Boolean function and compactly manipulate sets with large numbers of combinations. As we can roughly estimate that the number of combinations of free variables will not be large compared to the total number of combinations, we can suppress those combinations or variables which never appear, thus implementing it by one of the efficient variations of OBDD called *zero-suppressed BDD* (ZBDD) will be more suitable here. The data structure design and the construction method for ZBDD have been explicitly described in [25].

4 Concluding Remarks

In this paper, we propose an algebraic approach to the graph isomorphism problem by matching the representatives in advance. Some examples of graphs with representatives are provided in Section 4.1. The algorithm concisely utilizes matrix computation over the finite field $\mathbb{Z}_2$ in several aspects. Furthermore, this adoption enables the input to be directed graphs. By the nature of our heuristic flexibility on finding an invariant basis for the null space, we may explore more varieties along this line of investigation.

To carry out in practice the prime field arithmetic, there are some well-developed tools to put in use, such as the finite field linear algebra subroutines (FFLAS) and the finite field linear algebra package (FFPACK). FFLAS is a library based on the basic linear algebra subprograms (BLAS) interface for finite fields which performs highly optimized computation with automatically tuned numerical library (ATLAS) as a kernel; it can reduce the number of costly instructions and at the same time increase the locality of computation [12, 13].

An interesting open problem is how to theoretically randomize the guess based on such a scheme, or how to formulate the probability analysis on average and the relation between the number of variable combinations and that of graph automorphisms. Another problem is whether there exists any group theoretical transformation, such as the one in [6], under this formulation.

4.1 Graphs with Representatives

Besides considering purely the general graphs, our scheme can be particularly fitting and compatible under certain circumstances, in that finding corresponding representatives in the algorithm

tends to be a comparatively easy task. We name those graphs as the *feature-based* graphs, which signifies the property that there exists some distinguishable vertices in those graphs. Any particular type of graph can be conceptually viewed as a feature-based graph as long as we have already had some discrimination in the vertex set beforehand. Note that given any general type of graphs, there are some ways to transform them into vertex-colored graphs (which is a common step to initiate a GI algorithm but not within the scope of this paper), so that distinguishing features can reveal as well.

As an example of graphs with representatives, when detecting structural similarity of trees of XML documents, we can classify nodes into different groups by their location layers and label names. Similarly, we can also make some marks on social network graphs for each community cluster as well. Once having the ability to tell the differences between classes of vertices in advance, the cyclic basis is relatively uncomplicated to build.

As another example, *phylogenetic networks* have received much attention in the field of computational biology. A phylogenetic network is a directed acyclic graph used to present evolutionary events, such as recombination, hybridization, horizontal gene transfer or gene duplication and loss. If the prior mapping of species can be fixed, i.e., those specified leaf nodes (or species node) are separated into groups, then we can almost find representatives immediately. In [15], Grambette *et al.* encounter graph isomorphism testing in constructing the set of level- k generators which composes level- k phylogenetic networks. Since the hybrid nodes in each biconnected component can also become features, our heuristics can thus be applied.

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